Nokia Core Router Troubleshooting SOP for BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Node BLR-CORE-02

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Warning | **Alarm Code:** ALARM_1969

Description

This Standard Operating Procedure (SOP) is designed for the elite Telecom Network Architect to troubleshoot and resolve Routing Protocol Flap failures on Nokia Core Routers, specifically on Node ID BLR-CORE-02. The failure domain in question is BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, which is causing Alarm Code ALARM_1969 with a Warning severity. The underlying protocol mechanics that contribute to this failure domain include the dynamic nature of routing protocols, where frequent updates in the routing table can lead to instability in the routing process. Additionally, hardware mechanics such as high CPU utilization, memory congestion, and interface congestion can also exacerbate this issue. Furthermore, the specific configuration of the Nokia Core Router, including the BGP, OSPF, and ISIS protocols, can also play a role in this failure domain.

Pre-requisite Checks

- Verify that the Node BLR-CORE-02 is accessible and reachable via remote management interfaces.
- Check the redundant link status to ensure that it is operational and not contributing to the failure.
- Obtain the site location and physical access details to perform necessary troubleshooting and repairs.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 1, main routing table version 1 BGP send state: 0, BGP receive state: 0 BGP total number of peers: 2 BGP number of peers in established state: 0 ... , Description: This command displays a summary of the BGP table, including the total number of peers and the number of peers in the established state. It helps to identify if the BGP route churn is contributing to the failure.

Command: show ip ospf neighbor, Expected Output: OSPF neighbor table: Neighbor ID State Pri Dead Time Interface ... , Description: This command displays the OSPF neighbor table, which helps to identify if there are any OSPF adjacency loss issues contributing to the failure.
```

```
Command: show isis lsp detail, Expected Output: ISIS LSP detail: LSP ID Age Sequence Number Checksum ..., ... , Description: This command displays the ISIS LSP detail, which helps to identify if there is any ISIS LSP flooding contributing to the failure.
```

Resolution Steps

Step 1: Step: Verify BGP Configuration, Command: show bgp conf, Description: Verify the BGP configuration to ensure that it is correctly set up and not contributing to the failure.

Step 2: Step: Reset BGP Session, Command: clear bgp reset, Description: Reset the BGP session to clear any stale routing information and re-establish the BGP connection.

Step 3: Step: Check OSPF Configuration, Command: show ip ospf conf, Description: Verify the OSPF configuration to ensure that it is correctly set up and not contributing to the failure.

Step 4: Step: Reset OSPF Session, Command: clear ip ospf reset, Description: Reset the OSPF session to clear any stale routing information and re-establish the OSPF connection.

Step 5: Step: Verify ISIS Configuration, Command: show isis conf, Description: Verify the ISIS configuration to ensure that it is correctly set up and not contributing to the failure.

Step 6: Step: Reset ISIS Session, Command: clear isis reset, Description: Reset the ISIS session to clear any stale routing information and re-establish the ISIS connection.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 1, main routing table version 1 BGP send state: 0, BGP receive state: 0 BGP total number of peers: 2 BGP number of peers in established state: 2 ... , Description: Verify that the BGP route churn has been resolved and that the BGP table is stable.

Command: show ip ospf neighbor, Expected Output: OSPF neighbor table: Neighbor ID State Pri Dead Time Interface ... , Description: Verify that the OSPF adjacency loss has been resolved and that the OSPF neighbor table is stable.

Command: show isis lsp detail, Expected Output: ISIS LSP detail: LSP ID Age Sequence Number Checksum ..., ... , Description: Verify that the ISIS LSP flooding has been resolved and that the ISIS LSP detail is stable.
```

Roll-back Procedures

- Step: Restore BGP Configuration, Command: copy start bgp conf, Description: Restore the original BGP configuration to roll back any changes made during the troubleshooting process.

- Step: Restore OSPF Configuration, Command: copy start ip ospf conf, Description: Restore the original OSPF configuration to roll back any changes made during the troubleshooting process.

- Step: Restore ISIS Configuration, Command: copy start isis conf, Description: Restore the original ISIS configuration to roll back any changes made during the troubleshooting process.